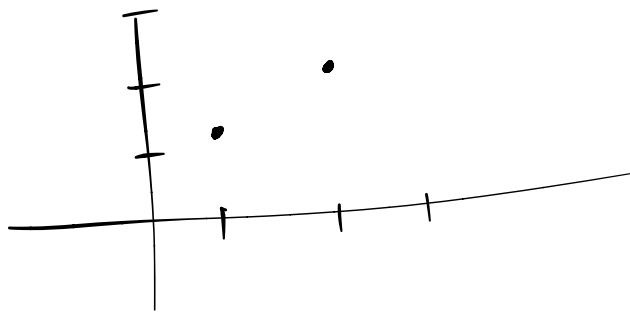


Lecture 13

Extra homework replaces 2nd lowest homework grade. Do all circled problems in additional problems for final.

Example least squares:

time (s)	velocity (m/s)
1	11
2	19
3	31



Model:

$$v = at + v_0$$

$$11 = a \cdot 1 + v_0$$

$$19 = a \cdot 2 + v_0$$

$$31 = a \cdot 3 + v_0$$

$$\begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} a \\ v_0 \end{bmatrix} = \begin{bmatrix} 11 \\ 19 \\ 31 \end{bmatrix}$$

$\tilde{X} \quad \quad \vec{\beta} \quad \quad \vec{y}$

$$\tilde{X}^T \tilde{X} \vec{\beta} = \tilde{X}^T \vec{y}$$
$$\tilde{X}^T \tilde{X} = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \end{bmatrix}$$
$$= \begin{bmatrix} 14 & 6 \\ 6 & 3 \end{bmatrix}$$

$$\tilde{X}^T \vec{y} = \begin{bmatrix} 142 \\ 61 \end{bmatrix}$$

$$\begin{bmatrix} 14 & 6 & | & 142 \\ 6 & 3 & | & 61 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & | & 10 \\ 0 & 1 & | & \frac{1}{3} \end{bmatrix}$$

$$v = 10t + \frac{1}{3}$$

To find projection of $\vec{y}$ onto $\text{ran } \tilde{X}$

$$\tilde{X} \vec{\beta} = \begin{bmatrix} 1 & | \\ 2 & | \\ 3 & | \end{bmatrix} \begin{bmatrix} 10 \\ \frac{1}{3} \end{bmatrix} = \begin{bmatrix} 10\frac{1}{3} \\ 20\frac{1}{3} \\ 30\frac{1}{3} \end{bmatrix}$$

Singular Value Decomposition

$$A = U \Sigma V^T$$

U - columns are the left singular vectors of A

Σ - diagonal matrix whose diagonal elements are singular values of A

V - columns are the right singular vectors of A

left/right singular vectors are the eigenvectors of AA^T/A^TA

singular values are the square roots of nonnegative eigenvalues of AA^T/A^TA .

If λ_i is an eigenvalue with v_i an eigenvector of $A^T A$ then

$$A^T A \vec{v}_i = \lambda_i \vec{v}_i$$

$$A A^T \underbrace{A \vec{v}_i}_{\text{left singular vector}} = \lambda_i \underbrace{A \vec{v}_i}_{\text{left singular vector}}$$

$$A = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -1 & 1 \\ -\frac{1}{2} & -\frac{1}{2} \end{bmatrix}$$

$$A^T A = \begin{bmatrix} \frac{3}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{3}{2} \end{bmatrix}$$

$$\det(A^T A - \lambda I) = 0$$

$$\lambda = 1, 2$$

singular values are $\sigma = \sqrt{1}, \sqrt{2}$.

right singular vectors are $\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$

Remember $A^T A, A A^T$ are symmetric so eigenvectors are orthogonal.

$$(A^T A - \lambda I) \vec{v} = 0$$

$$\lambda = 1 \quad \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \Rightarrow \vec{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ normalize to get } \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\frac{1}{2} v_1 - \frac{1}{2} v_2 = 0 \quad \text{Let } v_2 = 1.$$

$$\frac{1}{2} v_1 = \frac{1}{2} v_2$$

$$V = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

$$V^T V = I$$

$$U^T U = I$$

$$u_1 = A \vec{v}_1 = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -1 & -1 \\ -\frac{1}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \quad AA^T = \begin{bmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \\ 0 & 2 & 0 \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}$$

$$u_2 = A \vec{v}_2 = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -1 & -1 \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} 0 \\ -\sqrt{2} \\ 0 \end{bmatrix}$$

AA^T has eigenvalues 1, 2, 0

$$u_3 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & -\sqrt{2} & 0 \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & -1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$A = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -1 & -1 \\ -\frac{1}{2} & -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & -1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

principal component for $\sigma_1 > \sigma_2 > \dots$

$$A = U \Sigma V^T = \sigma_1 \vec{u}_1 \vec{v}_1^T + \sigma_2 \vec{u}_2 \vec{v}_2^T + \dots + \sigma_n \vec{u}_n \vec{v}_n^T$$

$$\begin{bmatrix} \vec{u}_1 & \vec{u}_2 & \vec{u}_3 \end{bmatrix} \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \vec{v}_1^T \\ \vec{v}_2^T \end{bmatrix}$$

$$\begin{bmatrix} \vec{u}_1 & \vec{u}_2 & \vec{u}_3 \end{bmatrix} \begin{bmatrix} \sigma_1 \vec{v}_1^T \\ \sigma_2 \vec{v}_2^T \\ 0 \end{bmatrix}$$

$\underbrace{\vec{u}_2^T \vec{v}_2}_{\text{inner product}}$

$$\sigma_1 \underbrace{\vec{u}_1 \vec{v}_1^T} + \sigma_2 \underbrace{\vec{u}_2 \vec{v}_2^T}$$

→ outer product produces a matrix

$$\vec{u} \vec{v}^T = (u_i v_j)$$

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 4 & 5 & 6 \\ 8 & 10 & 12 \\ 12 & 15 & 18 \end{bmatrix}$$

When reducing dimension of data store only largest k singular values and singular vectors.

Least squares

Given $A = QR$

Then

$$A^T A \vec{x} = A^T \vec{b}$$

$$R^T \cancel{Q^T Q}^I R \vec{x} = R^T Q^T \vec{b}$$

$$R^T R \vec{x} = R^T Q^T \vec{b}$$

$$R \vec{x} = Q^T \vec{b}$$

↑
upper triangular

$$\tilde{X} = \sum_{i=1}^n \sigma_i u_i v_i^T$$

↑
pick out most important features.

Ex: Predicting cost of houses